```cpp
#include <GL/glut.h>
#include <cmath>
#include <iostream>

void drawThickPixel(int x, int y) { glBegin(GL_POINTS); glVertex2i(x, y); glEnd(); }

void bresenhamLine(int x0, int y0, int x1, int y1) {
  int dx = abs(x1 - x0), sx = x0 < x1 ? 1 : -1;
  int dy = -abs(y1 - y0), sy = y0 < y1 ? 1 : -1;
  int err = dx + dy, e2;
  while (true) {
    drawThickPixel(x0, y0);
    if (x0 == x1 && y0 == y1) break;
    e2 = 2 * err;
    if (e2 >= dy) { err += dy; x0 += sx; }
    if (e2 <= dx) { err += dx; y0 += sy; }
  }
}

void drawCirclePoints(int xc, int yc, int x, int y) {
  drawThickPixel(xc + x, yc + y); drawThickPixel(xc - x, yc + y);
  drawThickPixel(xc + x, yc - y); drawThickPixel(xc - x, yc - y);
  drawThickPixel(xc + y, yc + x); drawThickPixel(xc - y, yc + x);
  drawThickPixel(xc + y, yc - x); drawThickPixel(xc - y, yc - x);
}
```

```
void bresenhamCircle(int xc, int yc, int r) {
   int x = 0, y = r, d = 3 - 2 * r;
   drawCirclePoints(xc, yc, x, y);
   while (y >= x) {
      x++;
      if (d > 0) { y--; d = d + 4 * (x - y) + 10; }
      else { d = d + 4 * x + 6; }
      drawCirclePoints(xc, yc, x, y);
   }
}

void display() {
   glClear(GL_COLOR_BUFFER_BIT);
   glPointSize(3.0);

   // 1. Brown Trunk
   glColor3f(0.35, 0.16, 0.14);
   for(int i = -3; i <= 3; i++) bresenhamLine(250 + i, 400, 250 + i, 50);

   // 2. Forest Green Leaves
   glColor3f(0.13, 0.55, 0.13);
   bresenhamLine(250, 150, 180, 220); bresenhamLine(180, 220, 220, 130);
bresenhamLine(220, 130, 250, 150);
   bresenhamLine(250, 250, 320, 320); bresenhamLine(320, 320, 280, 230);
bresenhamLine(280, 230, 250, 250);

   // 3. Autumn Petals
```

```
    float colors[8][3] = {
        {0.8, 0.2, 0.0}, {0.9, 0.4, 0.0}, {1.0, 0.6, 0.0}, {0.7, 0.1, 0.1},
        {0.6, 0.3, 0.0}, {0.9, 0.2, 0.2}, {0.8, 0.5, 0.2}, {1.0, 0.4, 0.2}
    };

    int cx = 250, cy = 450, pr = 40, dist = 75;
    for (int i = 0; i < 8; i++) {
        float angle = i * (45.0f * 3.14159f / 180.0f);
        glColor3f(colors[i][0], colors[i][1], colors[i][2]);
        bresenhamCircle(cx + cos(angle) * dist, cy + sin(angle) * dist, pr);
    }

    glColor3f(1.0, 0.8, 0.2); // Golden center
    bresenhamCircle(cx, cy, 35);
    glFlush();
}

int main(int argc, char** argv) {
    glutInit(&argc, argv); glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
    glutInitWindowSize(500, 700); glutCreateWindow("Earthy Flower");
    glMatrixMode(GL_PROJECTION); glLoadIdentity(); gluOrtho2D(0, 500, 0, 700);
    glClearColor(1.0, 1.0, 1.0, 1.0); glutDisplayFunc(display); glutMainLoop();
    return 0;
}
```